

COVER LETTER

21 April 2026

To the Editor / Programme Committee,

I am pleased to submit the manuscript titled *From Invisible to Cited: A Case Study of Generative Engine Optimisation for AI Search Recognition Among Indian Student Founders* for consideration in your journal or conference proceedings. This is an original work; it has not been published previously and is not under review elsewhere.

The manuscript addresses a gap that is both timely and practically significant. Generative Engine Optimisation (GEO) the systematic structuring of digital content so that AI-powered search platforms accurately retrieve and cite an entity is an emerging research area. Existing studies have been conducted almost exclusively in Western, well-funded startup contexts. This paper documents, to the authors' knowledge, among the first screenshot-evidenced single-case study of GEO implementation by a student entrepreneur at a regional Indian university, operating under a common personal name and a zero monetary budget.

The central finding that a structured sequence of SEO-compatible and GEO-oriented interventions was associated with coherent entity cluster formation in Google Search, independent entity node generation in Google Images, and entity-aware responses from all four major AI assistants tested (Perplexity, Gemini, Meta AI, ChatGPT) is documented through fourteen original screenshots and supported by a proposed three-stage SEO-GEO Integration Model.

The manuscript makes three specific contributions to the GEO literature: (1) an exploratory reference case for a structurally distinct practitioner population; (2) evidence suggesting that official institutional communication channels may function as accessible earned media pathways for student founders; and (3) a first direct platform-to-platform comparison of RAG versus parametric AI response granularity for a single, controlled entity across four simultaneous platforms.

The study is presented honestly as exploratory and self-referential; all limitations including the researcher-as-subject design, absence of a formal baseline, and single-rater scoring are disclosed in full in Section 7. The manuscript has been revised to meet international publication standards: claims are calibrated, all data are source-mapped in Appendix A, figures are captioned and numbered, tables are formatted, and the reference list is complete.

I believe this manuscript will be of interest to researchers in information retrieval, digital marketing, entrepreneurship in emerging markets, and the intersection of AI systems and online

identity. All evidence is available for reviewer inspection upon request. I welcome any editorial feedback.

Respectfully submitted,

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GEO CASE STUDY INDIAN STUDENT FOUNDER

From Invisible to Cited:

A Case Study of Generative Engine Optimisation for AI Search Recognition Among Indian Student Founders

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Manuscript submitted for review April 2026

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Author Note

Akash Kumar Singh is a postgraduate student enrolled in the Master of Computer Applications programme at Sarala Birla University (SBU), Ranchi, Jharkhand, India, and the Founder and Chief Executive Officer of Dilwado.com. This study is exploratory and self-referential in design: the author is simultaneously the researcher and the primary subject of investigation. This dual role is fully disclosed in Section 7 (Limitations) and has been considered in the framing and interpretation of all findings. The author reports no conflicts of interest beyond the researcher-as-subject role. No external funding was received. No institutional ethics approval was required, as all data consist of publicly indexed digital content generated by the author across platforms to which the author has provided consent. No personal data of third parties were collected, processed, or analysed at any stage. No human participants other than the author were involved. The fourteen screenshots constituting the primary empirical evidence are retained by the author and are available to reviewers upon request. Correspondence: meakash22dotin@gmail.com.

Abstract

The emergence of AI-powered generative search platforms including ChatGPT, Google Gemini, Meta AI, and Perplexity has introduced a dimension of digital visibility that conventional Search Engine Optimisation (SEO) frameworks were not designed to address. Generative Engine Optimisation (GEO) refers to the practice of structuring digital content and identity signals so that generative AI systems can accurately retrieve, synthesise, and cite a given entity in response to relevant queries. For student founders and early-stage startups operating under common Indian names, with limited domain authority and zero advertising budgets, this challenge is both practically urgent and largely undocumented in the scholarly literature.

This paper presents a documented single-case exploratory study of a student founder at Sarala Birla University (SBU), Ranchi, India, who implemented a structured, zero-budget sequence of SEO-compatible and GEO-oriented interventions over approximately seven months. The intervention sequence comprised: (a) keyword-aligned LinkedIn profile optimisation; (b) a structured founder entity page on the startup domain (Dilwado.com); (c) a static portfolio site with an identity-phrase domain slug; and (d) two officially published institutional posts from SBU documenting verifiable achievements. Evidence documented across fourteen original screenshots indicates that post-intervention queries produced a coherent first-page entity cluster in Google Search; entity-specific filter chips and at least three independent entity nodes in Google

Images; and entity-aware descriptions from all four major AI assistant platforms tested. A quantitative analysis of 28 query-platform trial combinations yielded a correct-entity retrieval rate of 100% within the tested sample, compared with 0% observed in informal pre-intervention checks.

Based on these observations, a three-stage SEO-GEO Integration Model is proposed: (1) Foundational Presence, (2) Entity Consolidation, and (3) Earned Authority Signals. To the authors' knowledge, this represents among the first screenshot-evidenced case studies of GEO implementation by a student entrepreneur at an Indian regional university, and it suggests that institutional earned media specifically, official university announcements of verified achievements may serve as an accessible GEO authority signal not previously identified in the literature.

Keywords: *generative engine optimisation; search engine optimisation; AI search visibility; entity disambiguation; digital identity; Indian startup ecosystem; student entrepreneurship; retrieval-augmented generation; knowledge graph*

1. Introduction

The nature of information search has undergone a qualitative transformation. Where conventional search engines return ranked lists of documents, AI-powered generative systems now synthesise direct answers by drawing on real-time retrieval and large-scale pre-training. Google AI Overviews, Perplexity's cited summaries, Gemini's contextual responses, and Meta AI's conversational panels represent a shift in information mediation whose commercial significance is increasingly documented. A fifteen-month analysis of 3,119 queries across 42 organisations found that organic click-through rates fell by approximately 65% and paid click-through rates by approximately 68% on queries where AI Overviews appeared, while entities cited within those Overviews received disproportionately higher engagement (Seer Interactive, 2025). Inclusion in an AI-generated answer is thus becoming as commercially consequential as first-page traditional search ranking.

This shift has given rise to Generative Engine Optimisation (GEO) the systematic practice of structuring digital content and identity signals to increase the probability that generative AI systems accurately retrieve, represent, and cite an entity when relevant queries are posed. GEO is not simply SEO adapted to new systems; it involves different signals, different architectural constraints, and different optimisation principles. Aggarwal et al. (2024) formally introduced and benchmarked GEO, finding that structured, citation-rich, fact-dense content was associated with up to 40% higher citation rates in generative engine outputs compared with unoptimised equivalents. Sharma (2026), examining 112 startups across 2,240 queries, found that traditional SEO metrics predicted Perplexity visibility ($r = 0.319$, $p < 0.001$) but showed no statistically significant association with ChatGPT discoverability a finding consistent with a fundamental architectural divergence between the two disciplines. Chen et al. (2025) proposed that AI search systems systematically weight earned, third-party content over brand-owned material, outlining a GEO agenda fundamentally distinct from the backlink-centric logic of traditional SEO.

What the GEO literature has not examined is how these principles apply to practitioners operating outside Western media ecosystems and without funding. India is estimated to have the world's third-largest startup count (NASSCOM, 2024), with a substantial proportion of ventures originating at regional universities. These founders frequently operate under common names 'Akash Kumar Singh,' for example, is shared by tens of thousands of individuals across Indian states with no advertising budget and recognition structures mediated by institutions (universities, government schemes, technical competitions) rather than technology press. Whether GEO, as theorised and empirically tested to date, is actionable for this population has not previously been investigated.

This paper addresses that gap through a single, carefully documented case. The subject is an MCA student and startup founder at Sarala Birla University, Ranchi, who implemented a structured, zero-budget intervention sequence over approximately seven months. Post-intervention AI search visibility is documented through fourteen original screenshots across Google Search, Google Images, ChatGPT, Google Gemini, Meta AI, and Perplexity. The paper

makes three specific, bounded contributions: (1) to the authors' knowledge, it presents among the first screenshot-evidenced documentation of GEO implementation by a student entrepreneur at a regional Indian university; (2) it suggests that official university communication channels may function as effective GEO earned media signals, a pathway not previously prominent in the GEO literature; and (3) it proposes a three-stage SEO-GEO Integration Model grounded in the observed intervention sequence and structured to be reproducible by practitioners under comparable resource constraints.

Three research questions frame the empirical analysis:

- **RQ1:** Following a structured sequence of zero-budget SEO-compatible and GEO-oriented interventions, how do Google Search and Google Images respond to identity-related queries involving the subject?
- **RQ2:** Do major AI assistants (ChatGPT, Google Gemini, Meta AI, Perplexity) produce entity-aware descriptions that accurately associate the subject with their startup and university affiliation?
- **RQ3:** Can the observed intervention sequence be conceptualised as a reproducible model applicable to other Indian student startup founders operating under similar resource constraints?

The study is exploratory and single-case in design. It does not seek statistical generalisation; all findings are framed as associative rather than causal. Epistemic limitations are acknowledged in full in Section 7, and claims are calibrated throughout to reflect what this evidence can and cannot establish.

2. Related Work

2.1 From SEO to GEO: A Structural Transition

Search Engine Optimisation has been the dominant paradigm for managing online visibility since the late 1990s, built around signals such as keyword placement, inbound link authority, technical site quality, and user engagement metrics (Beel et al., 2010; Baye et al., 2016). The field is mature and extensively documented in the practitioner and academic literature.

The emergence of generative AI as an information intermediary introduces a structural discontinuity. Generative systems synthesise answers rather than returning documents, either by retrieving content at inference time through Retrieval-Augmented Generation (RAG; Lewis et al., 2020) or by drawing on patterns encoded during large-scale pre-training (parametric generation). Aggarwal et al. (2024) formalised this distinction, introducing GEO and the GEO-Bench benchmark, and demonstrating that structured, citation-rich, fact-dense content achieved meaningfully higher citation rates across nine content domains.

The most consequential empirical finding for the present study comes from Sharma (2026, arXiv preprint), who examined 112 Product Hunt startups across 2,240 queries. Referring domains correlated significantly with Perplexity visibility ($r = 0.319$, $p < 0.001$), consistent with Perplexity's RAG architecture. No tested SEO metric showed a statistically significant association with ChatGPT discoverability suggesting the mechanisms underlying GEO differ substantially across AI architectures. Chen et al. (2025, arXiv preprint) identified three systematic tendencies: AI systems appear to prefer earned, third-party content over brand-owned material; response quality varies across platforms; and AI systems are more sensitive to query wording than traditional search engines.

2.2 Entity Disambiguation and Knowledge Graphs

A dimension of GEO that remains undertheorised is entity disambiguation, the process by which search and AI systems determine that references across multiple independent sources pertain to the same real-world individual or organisation. Google's Knowledge Graph, introduced in 2012 (Singhal, 2012) and central to both Search Knowledge Panels and AI-mediated responses, constructs entity profiles by integrating structured data, cross-domain consistency of named attributes, authoritative source signals, and earned media corroboration (Noy et al., 2019). An entity with consistent cross-domain naming and independently corroborated attributes is more likely to be included in the Knowledge Graph and, by extension, more accurately represented in AI-generated answers.

For Indian founders with common names, this challenge is particularly acute. Without deliberate disambiguation signals, AI systems queried for 'Akash Kumar Singh' are liable to return either a generic disambiguation notice or a conflation with another individual. Volpini et al. (2026, arXiv preprint) introduced the concept of structured linked data as a persistent entity memory layer for

agent-orchestrated retrieval systems, arguing that machine-readable, cross-domain entity representations enhance accessibility across AI architectures a framework that directly informs the Entity Consolidation stage of the three-stage model proposed in Section 5.

2.3 Research Gaps Addressed by This Study

Table 1 maps specific gaps in the existing GEO literature against the contributions of this study.

Table 1

Gaps in Existing GEO Literature and This Study's Contributions

Gap in Existing Literature	Why It Matters	This Study's Response	Nature of Contribution
GEO research conducted almost entirely in Western, well-funded startup contexts	Strategies and constraints differ in emerging markets; Product Hunt findings may not transfer	Documents a zero-budget case at a regional Indian university with a common-name founder	Exploratory reference case for a structurally distinct context
No prior study documents institutional earned media as a GEO signal	Chen et al. (2025) confirm earned media matters, but examples involve press coverage inaccessible to student founders	Observes official university announcements appearing in AI responses across four platforms	Suggests institutional channels as a potential earned media pathway; requires multi-case confirmation
GEO theory not applied at individual founder entity level	Common names introduce entity ambiguity not present for uniquely named products	Examines GEO at individual entity level with cross-platform consistency and institutional corroboration	Shows common-name disambiguation consistent with entity consolidation principles
No prior study compares RAG vs. parametric AI platforms using a single controlled entity	Sharma (2026) uses aggregate data; single-entity comparison allows direct platform observation	Compares ChatGPT, Gemini, Meta AI, and Perplexity responses for one case entity	Reveals RAG-granular/hybrid-comprehensive/parametric-summary pattern for a single entity

Note. GEO = Generative Engine Optimisation; RAG = Retrieval-Augmented Generation. Gap identification based on review of Aggarwal et al. (2024), Sharma (2026), Chen et al. (2025), and Volpini et al. (2026).

3. Methodology

3.1 Research Design and Epistemological Position

This study adopts a single-case exploratory design following the case study methodology described by Yin (2018). The design is appropriate for three reasons: (a) the combination of characteristics student founder, common personal name, regional Indian university, zero operational budget, EdTech domain represents a context structurally new to the GEO literature that cannot be studied through existing aggregate datasets; (b) the primary aim is to document a mechanism and theorise a process rather than estimate the frequency or magnitude of an effect; and (c) fourteen original screenshots collected across six platforms under multiple query conditions provide a substantive evidentiary basis for a single-case observational study in this nascent field.

This design has well-known limitations: it does not permit statistical generalisation; the researcher-as-subject role introduces potential for confirmation bias; and pre-intervention data were not systematically logged. All of these limitations are disclosed in Section 7. The study is presented as an exploratory reference case, not a definitive empirical test, and all claims are calibrated accordingly. Observed patterns should be treated as hypothesis-generating rather than hypothesis-confirming.

3.2 Subject and Context

The subject is an MCA student (2024–present) at Sarala Birla University (SBU), Ranchi, Jharkhand, India, with a prior BCA from the same institution (2021–2024) and a self-reported cumulative CGPA of 8.2. He is the founder and CEO of Dilwado.com, established in February 2020 an AICTE-approved internship and training platform that, per author-reported data on public profiles, serves over 10,000 registered students on a React/Node.js/MongoDB stack. Additional credentials verifiable through publicly indexed sources at the time of observation include: selection as a Google Student Ambassador; STEM Mentor appointment at Agastya International Foundation (since December 2025); organisation and branding lead for Tech Pragati 2026, SBU's annual technical festival; invited speaker at ICACA-2026; IIC Convener and Placement Coordinator at SBU; and a YouTube channel with over 400,000 subscribers and 13.7 million total views (author-reported, publicly viewable). All credentials are mapped to their public sources in Appendix A. No paid advertising, purchased backlinks, sponsored content, or commercial SEO services were used at any point during the study period.

3.3 Intervention Sequence

Interventions were implemented over approximately seven months prior to the observation window and classified into two functional categories: SEO-compatible actions and GEO-oriented entity and authority signals. In practice, these categories are interdependent. Table 2 provides a structured summary of all interventions, their mechanisms, and their theoretical grounding.

Table 2*Intervention Summary: Actions, Categories, and Theoretical Mechanisms*

Action	Category	What Was Done	Theoretical Mechanism
LinkedIn optimisation	SEO-comp atible	URL slug revised to include identity phrase; headline set to 'MCA Student Tech Entrepreneur Founder & CEO of Dilwado.com (AICTE-Verified)'; About section rewritten with structured, factual identity narrative; Experience updated for cross-platform consistency	LinkedIn's high domain authority causes its pages to rank prominently for personal name queries. Keyword alignment in URL, headline, and body increases relevance for identity queries (Beel et al., 2010).
Dilwado.com founder entity page	SEO + GEO	Long-form page titled 'Who is Akash Kumar Singh Tech?'; identity phrase in URL, H1, and body; fact-dense: MCA enrolment, CGPA 8.2, founding date Feb 2020, AICTE approval, 10,000+ students, tech stack	Fact-dense structured content on an established domain creates a high-probability GEO citation surface. Specificity of verifiable claims increases RAG retrieval fidelity (Aggarwal et al., 2024).
Static portfolio site (Netlify)	SEO + GEO	Static site at akash-kumar-singh-tech.netlify.app; mirrors core identity facts; identity phrase embedded in subdomain string; consistent professional photograph across all properties	Third independently indexed domain reinforces cross-domain entity signals. Domain-slug strategy may provide a Perplexity-accessible image surface (Volpini et al., 2026).
Cross-platform visual consistency	GEO – entity	Identical professional photograph deployed on LinkedIn, Dilwado.com entity page, and Netlify portfolio	Consistent visual signals across independent domains may support cross-platform photo entity linking, enabling entity chip generation in Google Images (Noy et al., 2019).
Uniform role nomenclature	GEO – entity	Identical role titles across all properties: 'Tech Entrepreneur', 'Founder & CEO of Dilwado.com',	Consistent attribute naming is a material input to Knowledge Graph entity construction; inconsistency impairs cross-source entity

Action	Category	What Was Done	Theoretical Mechanism
		'MCA student at Sarala Birla University'	merging (Noy et al., 2019; Volpini et al., 2026).
SBU – Google Student Ambassador post	GEO – earned media	Official SBU Facebook post naming subject as Google Student Ambassador; 40+ reactions; indexed approximately 7 months prior to observation window	Earned, third-party institutional content carries systematically higher weight in AI retrieval than brand-owned content. Institutional source authority is critical for GEO (Chen et al., 2025).
SBU – AIRAVATA hackathon post	GEO – earned media	Official SBU Facebook post naming subject as leader of Team AIRAVATA; 50+ reactions; indexed approximately 7 months prior to observation window	Second independent institutional authority signal from the same source; corroborating earned media may strengthen entity credibility for AI synthesis (Chen et al., 2025).

Note. SEO = Search Engine Optimisation; GEO = Generative Engine Optimisation; RAG = Retrieval-Augmented Generation; AICTE = All India Council for Technical Education; IIC = Institution's Innovation Council; SBU = Sarala Birla University.

3.4 Baseline and Retrospective Comparison

The absence of formally logged pre-intervention data is the most consequential methodological limitation of this study and is acknowledged as such in Section 7. Observations were conducted on at least two physical devices (a personal laptop and a smartphone) under more than one Google account, and repeated on multiple days to reduce the influence of temporary personalisation effects. The Google Search result at Position 1 in Figure 1 a different individual's LinkedIn profile serves as a structural marker of the pre-intervention disambiguation problem: it demonstrates concretely that the name is shared by multiple indexed individuals and that query resolution was not entity-specific prior to interventions. Pre-intervention informal checks consistently returned either no results pertaining to the subject or results indistinguishable from other individuals sharing the name. These observations are reported descriptively in Table 4 and explicitly labelled as informal. No causal inference is drawn from the before–after comparison.

3.5 Observation Procedure

Observations were conducted in April 2026, after all interventions had been live and presumed fully indexed for several weeks. Across all platforms, a total of four query forms were tested for each of six observation surfaces (Google Search, Google Images, Perplexity, Gemini, Meta AI, ChatGPT), yielding 24 primary query-surface combinations and 28 documented trials (some

queries tested on multiple devices). For AI assistant platforms, queries were submitted independently, repeated on different days, and responses were cross-checked against publicly available information for factual accuracy. Screenshots were captured and retained for every configuration tested. Personalisation was not fully controlled; this limitation is acknowledged in Section 7. It is further noted that all granularity scores in Table 5 were assigned by a single rater (the author); independent inter-rater validation has not been conducted, and scores should be treated as indicative rather than definitive.

Four query forms were tested: (a) a role-qualified name query ('mca student akash kumar singh'), (b) a domain-qualified query ('akash kumar singh tech'), (c) a geographic variant ('in ranchi'), and (d) a second geographic variant ('in jharkhand'). This design permitted observation of entity disambiguation under varying query specificity conditions.

4. Results

Results are presented across four subsections: Google Search (Section 4.1), Google Images (Section 4.2), AI assistant responses (Section 4.3), and quantitative summary tables (Section 4.4). All figures are original screenshots documented in April 2026 and are retained by the author for reviewer inspection upon request.

4.1 Google Search: Entity-Focused First-Page Cluster (RQ1)

Figure 1 presents the Google Search results page for the query 'mca student akash kumar singh', simultaneously illustrating the post-intervention outcome and the baseline disambiguation problem the intervention sequence was designed to address.

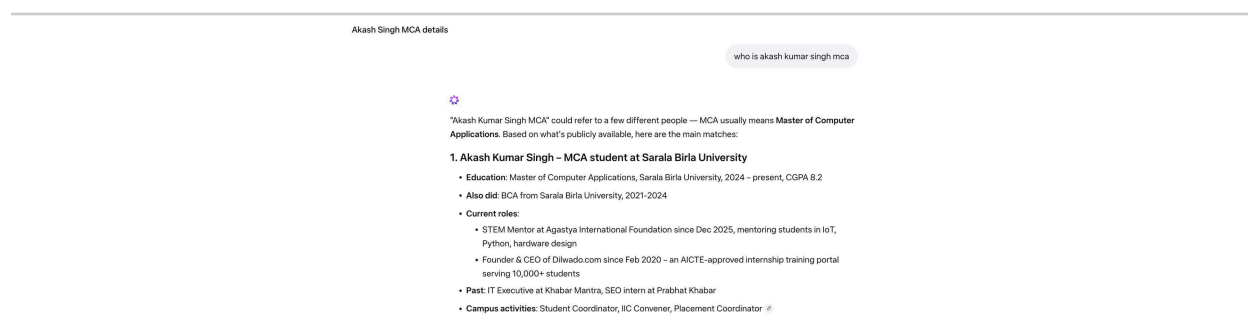


Figure 1. Google Search results for 'mca student akash kumar singh' (April 2026). Result 1 belongs to a different individual (PES University), illustrating the pre-intervention disambiguation challenge. Results 2–5 form the post-intervention entity cluster, drawn from three structurally distinct domain types: LinkedIn (Result 2), Dilwado.com (Result 3), and two SBU institutional Facebook posts (Results 4–5). Screenshot retained by author; available upon request.

Result 1 is a LinkedIn profile for a different individual an MCA student at PES University. This result makes the entity disambiguation problem concrete: prior to interventions, the subject's profile was indistinguishable from this result in standard search queries. Results 2 through 5 form the post-intervention cluster: the subject's LinkedIn profile; the Dilwado.com entity page ('Who is Akash Kumar Singh Tech?'); the SBU Facebook post announcing the Google Student Ambassador selection (40+ reactions, indexed approximately seven months prior); and the SBU Facebook post recognising Team AIRAVATA (50+ reactions).

The significance of this result lies not in the ranking of individual pages but in the emergence of a coherent, cross-sourced entity cluster. Four consecutive results pertain exclusively to the subject, drawn from three structurally distinct domain types: a professional networking platform (LinkedIn), the subject's own startup domain (Dilwado.com), and two posts from an independently operated institutional account (SBU Facebook). This cross-source consistency is consistent with what Noy et al. (2019) characterise as the input condition for Knowledge Graph entity construction and what Volpini et al. (2026) identify as a prerequisite for agent-orchestrated retrieval.

4.2 Google Images: Entity Node Formation and Visual Disambiguation (RQ1)

The Google Images evidence is, in several respects, more theoretically informative than the text search evidence, because it makes entity-level signal consolidation directly visible through filter chips and entity side panels. Figures 2 through 7 document this process across six distinct result states.

4.2.1 Filter Chips and Primary Identity Label

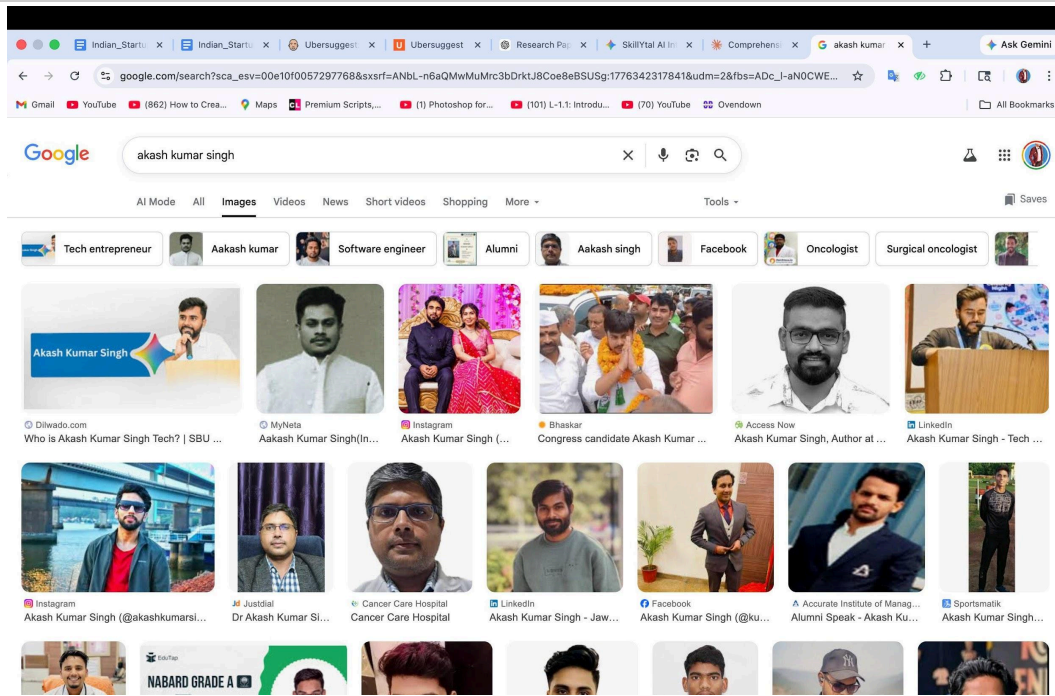


Figure 2. Google Images filter chips for 'akash kumar singh' (April 2026). 'Tech entrepreneur' appears as the first chip, encoding the consistently deployed role phrase as a categorical entity label derived from cross-domain signal analysis. Screenshot retained by author.

The first filter chip generated is 'Tech entrepreneur' the exact role phrase deployed consistently across the subject's LinkedIn headline, the Dilwado.com entity page, and the Netlify portfolio. This is consistent with Google's entity subsystem reading consistent signals across independent domains and encoding them as categorical entity labels a subsystem distinct from the standard search ranking algorithm.

4.2.2 Top Result Row and Three-Domain Cluster

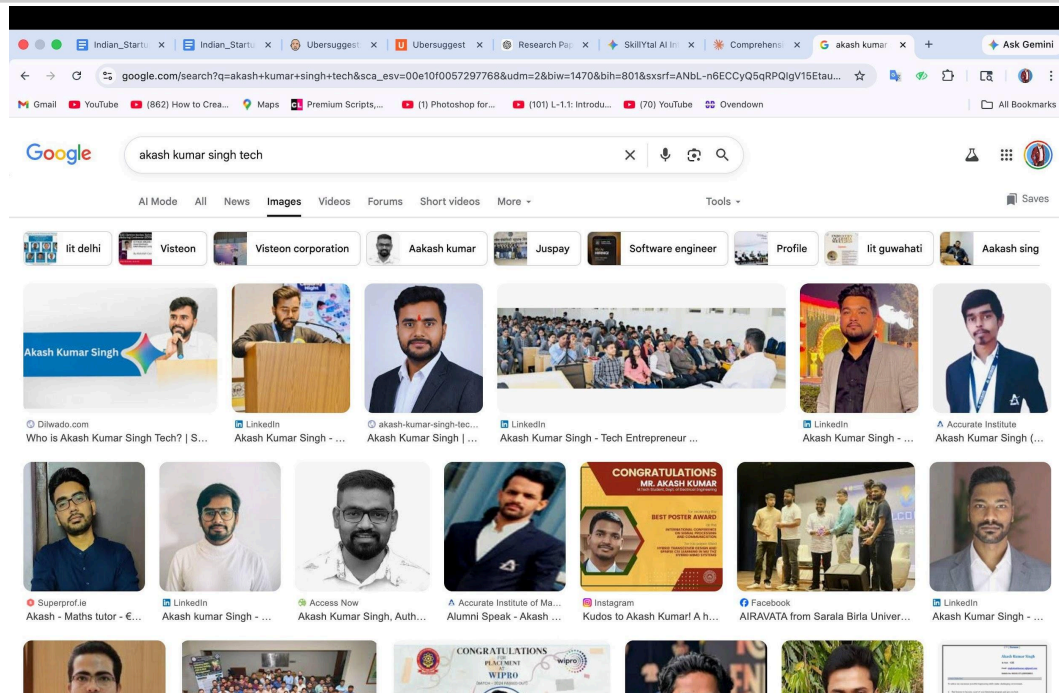


Figure 3. Google Images top results for 'akash kumar singh tech' (April 2026). Three intervention domains (Dilwado.com, LinkedIn, akash-kumar-singh-tech.netlify.app) appear as a coherent cluster in the top positions, retrieved as a group consistent with cross-domain entity signal consolidation. Screenshot retained by author.

The top positions of the Google Images results for 'akash kumar singh tech' are occupied by the three intervention domains. Three independently hosted domains, each carrying the same identity phrase in their content or URL, are retrieved as a coherent group consistent with cross-domain entity signal consolidation as described by Volpini et al. (2026).

4.2.3 LinkedIn Entity Node

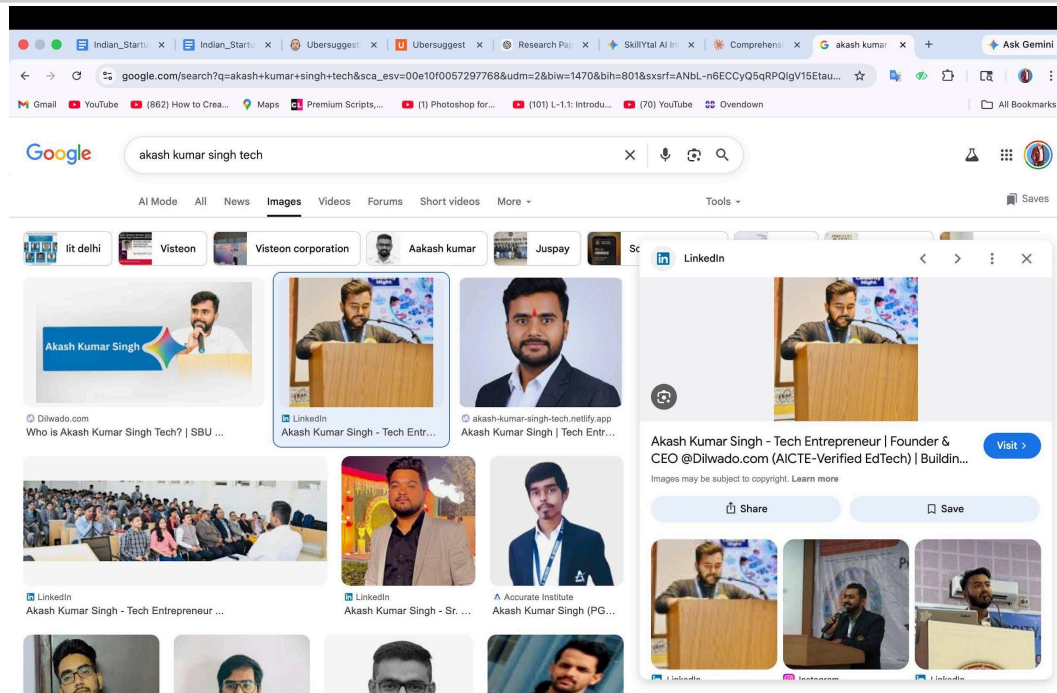


Figure 4. Google Images LinkedIn entity side panel for 'akash kumar singh tech' (April 2026). The panel displays the GEO-optimised headline and draws thumbnail images from multiple platform sources, suggesting cross-platform photo associations extending beyond the three directly optimised domains. Screenshot retained by author.

The LinkedIn side panel displays the GEO-optimised headline and draws thumbnail images from LinkedIn posts, an Instagram image, and two Facebook posts. This is consistent with Google's entity graph extending cross-platform photo associations beyond the three directly optimised domains to include at least four additional source platforms.

4.2.4 Netlify Portfolio Entity Node

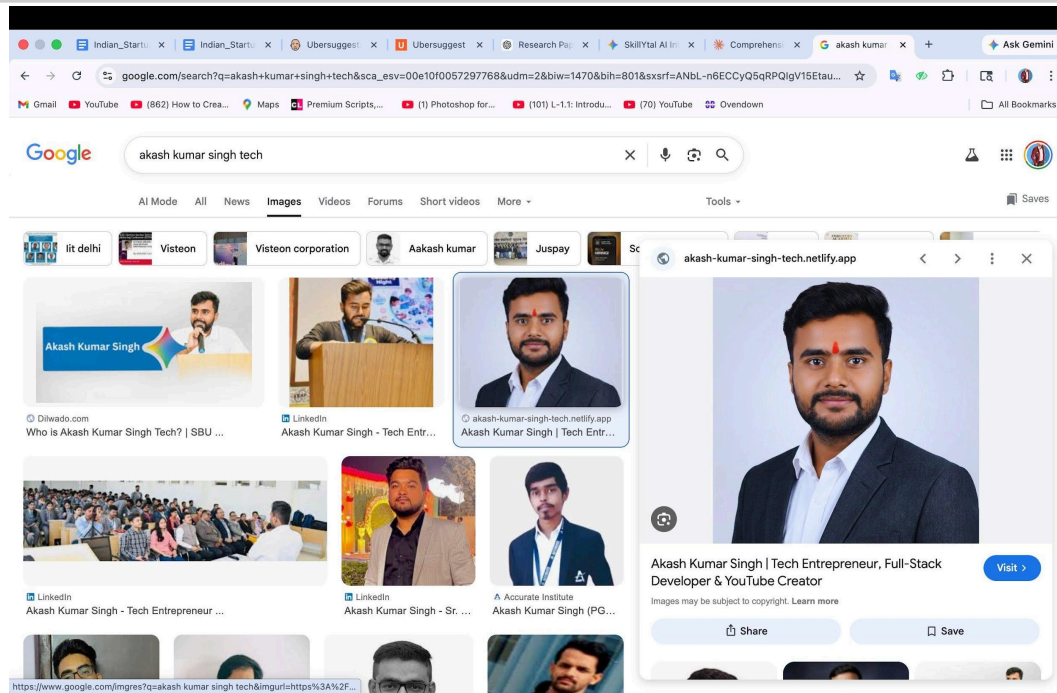


Figure 5. Google Images Netlify portfolio entity node for 'akash kumar singh tech' (April 2026). The identity-phrase domain slug strategy resulted in the portfolio being indexed as an independent named entity node, subsequently cited by Perplexity as a primary source across all three geographic query variants. Screenshot retained by author.

The domain-slug strategy embedding the identity phrase in the subdomain string resulted in the portfolio site at akash-kumar-singh-tech.netlify.app being indexed as an independent, named entity node. This domain is subsequently cited as a primary source by Perplexity across all three query variants tested (Figures 8, 12, and 13).

4.2.5 Dilwado.com Entity Node and Cross-Platform Photo Associations

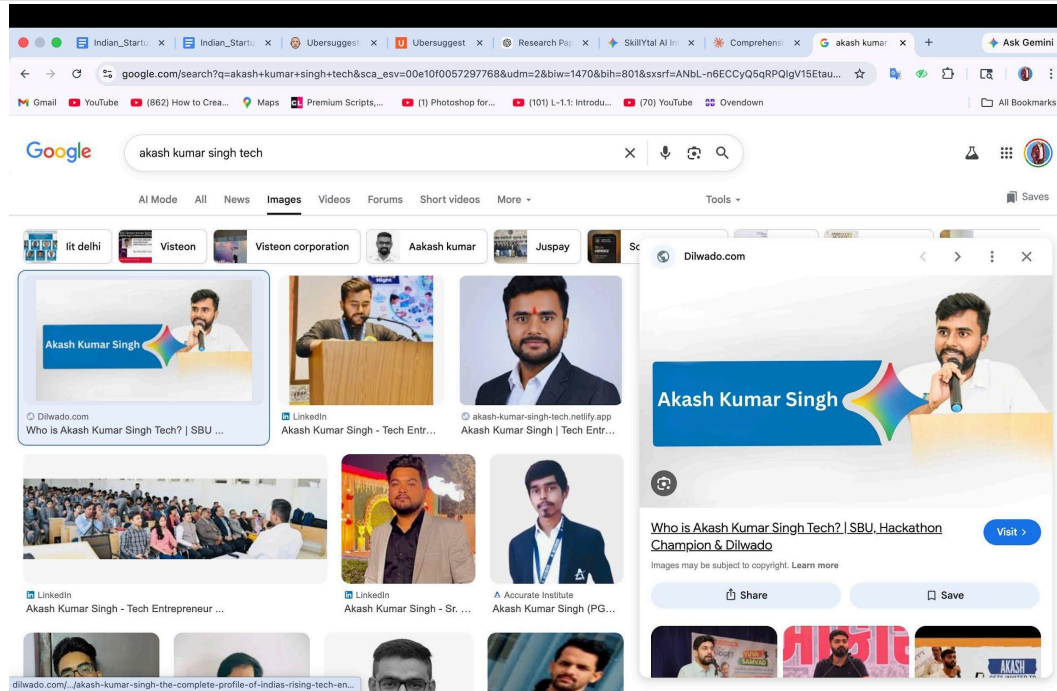


Figure 6. Google Images Dilwado.com entity node (April 2026). Screenshot retained by author.

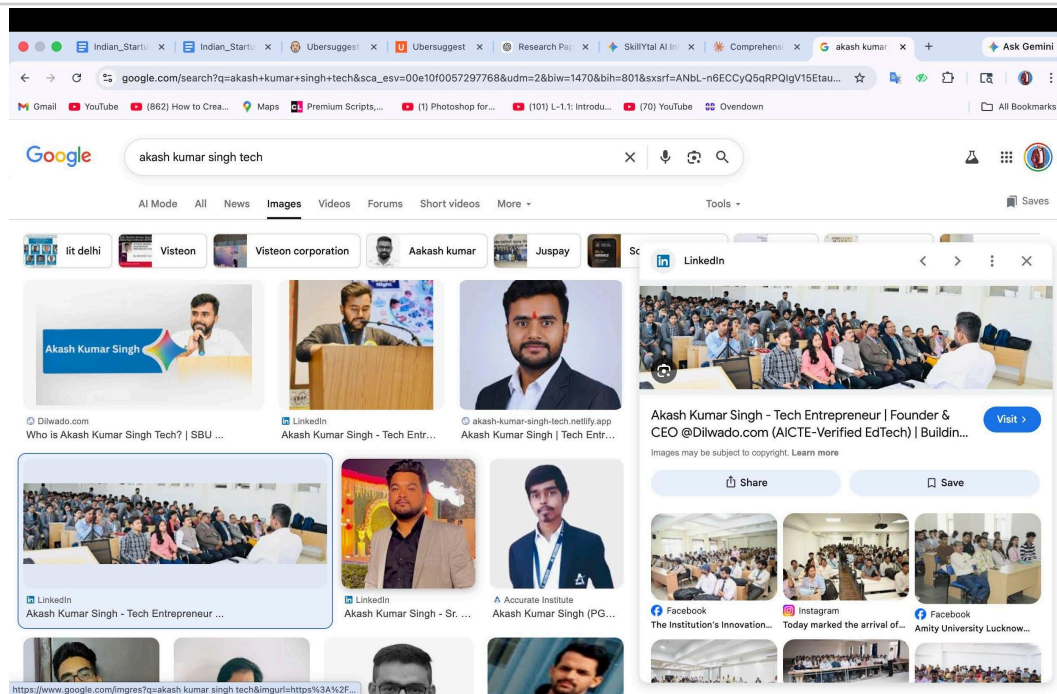


Figure 7. Google Images LinkedIn event photograph node with cross-platform associations (April 2026). Thumbnail gallery draws from LinkedIn, Facebook (IIC event), Instagram, and Facebook (Amity University) four distinct platform sources contributing to a single entity node, consistent with Knowledge Graph entity inclusion as described by Noy et al. (2019). Screenshot retained by author.

Figure 7 provides the clearest evidence in the dataset of cross-platform entity consolidation. The thumbnail gallery draws images from LinkedIn, Facebook (IIC event), Instagram, and Facebook (Amity University) four distinct platform sources contributing to a single entity node. Taken together, Figures 2 through 7 document cross-platform entity consolidation extending beyond the three directly optimised domains, drawing content from at least six distinct platform sources. The 'Tech entrepreneur' filter chip (Figure 2) represents the most compressed expression of this consolidation.

4.3 AI Assistant Responses (RQ2)

Figures 8 through 14 document AI assistant responses across four platforms and six distinct queries. Table 3 provides a comparative analysis of platform architecture and observed response behaviour.

Table 3

AI Platform Architecture and Observed Response Behaviour Across Four Platforms

Platform	Architecture	Key Data Observed (This Study)	Interpretive Note
Perplexity AI	RAG	CGPA 8.2; MCA 2024–present; BCA 2021–2024; Dilwado.com Feb 2020; 10,000+ students (author-reported); AICTE-approved; STEM Mentor; IIC Convener; Placement Coordinator; photo carousel from Netlify domain; 4 referring domains cited (Ranchi query)	RAG engines retrieve live indexed content at inference time. GEO-optimised structured content appears directly ingested and synthesised. Consistent with Sharma (2026) $r = 0.319$ finding.
Meta AI	Hybrid RAG + parametric	React/Node.js/MongoDB stack; Anveshna National Competition detail; 300+ students in Agile programmes (author-reported); AICTE-approved; campus roles; Netlify domain cited	Hybrid architecture retrieves structured profile data with notable fidelity. Tech stack and competition details originate exclusively from GEO-optimised profiles.
Google Gemini	Hybrid (Google index + LLM)	Student leader at SBU; Google Student Ambassador; Tech Pragati 2026 organiser; ICACA-2026 speaker; Dilwado.com 10,000+ students; YouTube 400,000+ subscribers and 13.7M views (publicly viewable)	Retrieves beyond directly optimised properties, including institutional event pages and YouTube channel data not part of the intervention sequence.

Platform	Architecture	Key Data Observed (This Study)	Interpretive Note
ChatGPT (GPT-4o)	Primarily parametric	Accurate summary: associated subject with SBU, Dilwado.com, and Google Student Ambassador role. Less granular than RAG-based platforms.	No statistically significant association between SEO metrics and ChatGPT discoverability (Sharma, 2026). Accuracy attributable to indexed content density in training data.

Note. RAG = Retrieval-Augmented Generation. Architecture classifications based on publicly available platform documentation. GEO sensitivity assessments are interpretive, derived from observed response granularity. Metrics marked 'author-reported' are drawn from the subject's public profiles and have not been independently verified.

4.3.1 Perplexity AI 'who is akash kumar singh mca' (Figure 8)

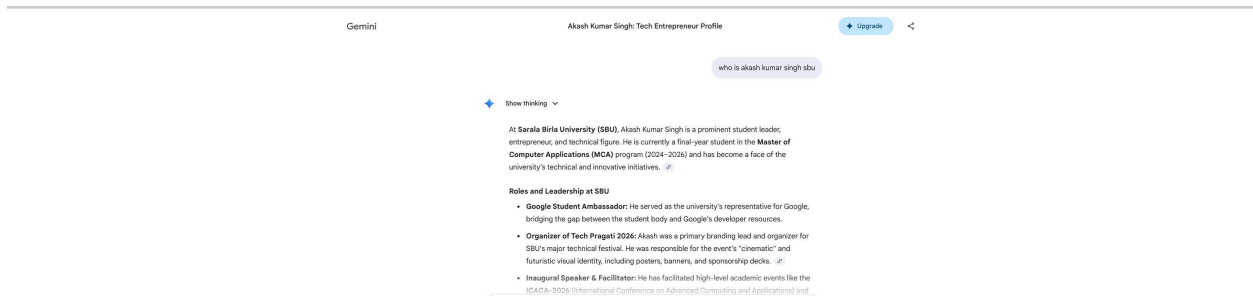


Figure 8. Perplexity AI response for 'who is akash kumar singh mca' (April 2026). Response is structured as a named entity profile. CGPA 8.2, BCA/MCA timelines, STEM Mentor role, Dilwado.com founding date, and three campus roles are accurately retrieved. Primary source cited: akash-kumar-singh-tech.netlify.app. Screenshot retained by author.

The response is headed 'Akash Kumar Singh – MCA student at Sarala Birla University' and retrieves CGPA 8.2, MCA 2024–present, BCA 2021–2024, STEM Mentor role at Agastya International Foundation since December 2025, Dilwado.com founding date and student count, past roles, and three SBU campus positions. Their accurate reproduction is consistent with Perplexity retrieving and synthesising the designed content rather than generic web data about the name.

4.3.2 Google Gemini 'who is akash kumar singh sbu' (Figure 9)

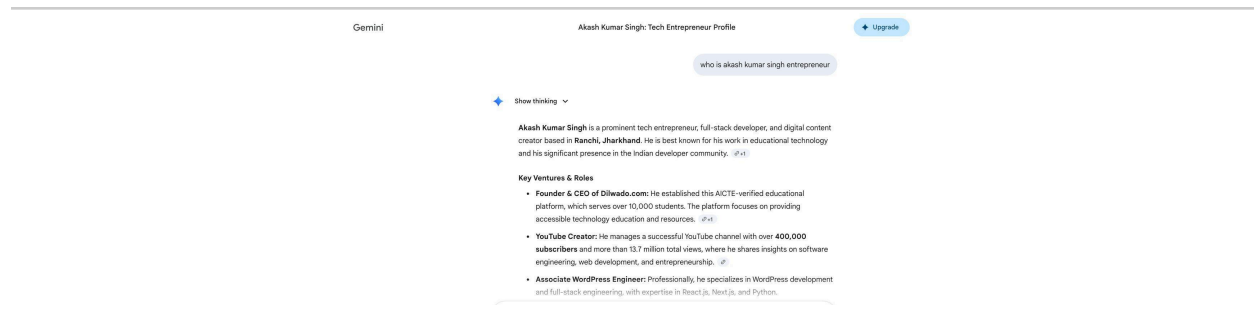


Figure 9. Google Gemini response for 'who is akash kumar singh sbu' (April 2026). Gemini labels the session 'Akash Kumar Singh: Tech Entrepreneur Profile', correctly attributing the Google Student Ambassador role, Tech Pragati 2026 organisation responsibilities, and the ICACA-2026 speaking invitation attributes not explicitly present in the three directly optimised properties. Screenshot retained by author.

Gemini labels the conversation 'Akash Kumar Singh: Tech Entrepreneur Profile'. The response correctly attributes the Google Student Ambassador role, Tech Pragati 2026 organisation responsibilities, and the ICACA-2026 speaking invitation. Notably, Tech Pragati 2026 and ICACA-2026 roles are not explicitly present in the three directly optimised properties, suggesting Gemini is retrieving from institutional event pages consistent with AI entity representation not being bounded by what a founder deliberately optimises.

4.3.3 Google Gemini 'who is akash kumar singh entrepreneur' (Figure 10)

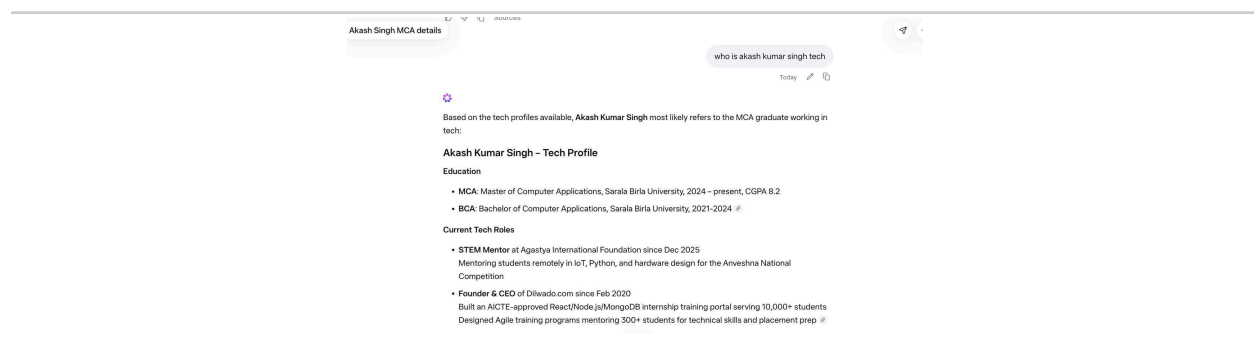


Figure 10. Google Gemini response for 'who is akash kumar singh entrepreneur' (April 2026). YouTube metrics (400,000+ subscribers, 13.7M views, publicly viewable) are drawn from the public YouTube channel a property not part of the intervention sequence confirming Gemini's retrieval extends across the subject's broader public digital footprint. Screenshot retained by author.

The response accurately describes Dilwado.com as an AICTE-verified platform serving over 10,000 students (author-reported) and attributes the YouTube channel's 400,000 subscribers and 13.7 million total views drawn from a property not part of the intervention sequence. This confirms that Gemini's retrieval is not bounded by the directly optimised properties.

4.3.4 Meta AI 'who is akash kumar singh tech' (Figure 11)

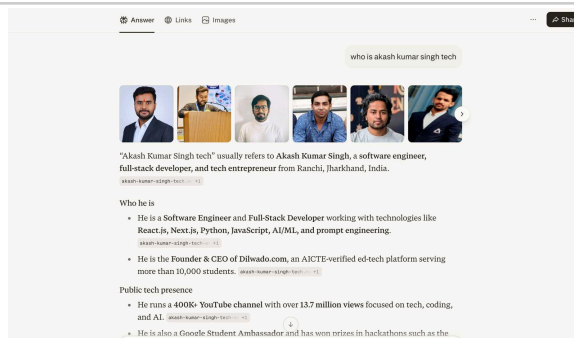


Figure 11. Meta AI response for 'who is akash kumar singh tech' (April 2026). Structured as 'Akash Kumar Singh – Tech Profile'. The Anveshna National Competition detail and 300+ Agile training students figure (author-reported) are among the most specific data points across all AI responses in this study; Netlify domain cited as source. Screenshot retained by author.

Meta AI produces the most granular single-query response of any platform tested for this query form. The Anveshna National Competition detail and the 300+ students mentored through Agile training programmes (author-reported) are among the most specific data points across all AI responses. Their presence, with the Netlify portfolio cited as source, is consistent with the designed content being retrieved and reproduced with high fidelity.

4.3.5 Perplexity AI Photo Carousel and Geographic Queries (Figures 12–13)

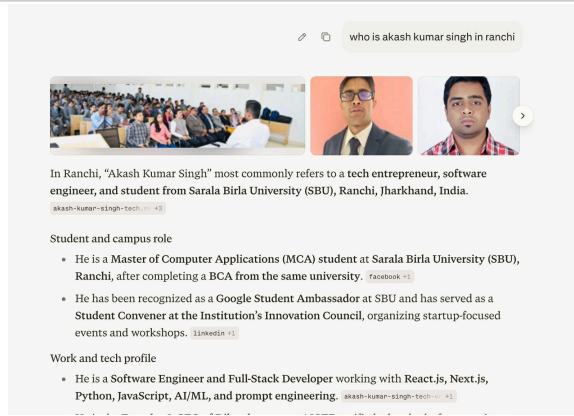


Figure 12. Perplexity AI response for 'who is akash kumar singh tech' showing a photo carousel sourced from akash-kumar-singh-tech.netlify.app (April 2026). This is the only instance in the dataset of a photo carousel in an AI assistant response, consistent with the domain-slug strategy producing an AI-mediated visual entity representation. Screenshot retained by author.

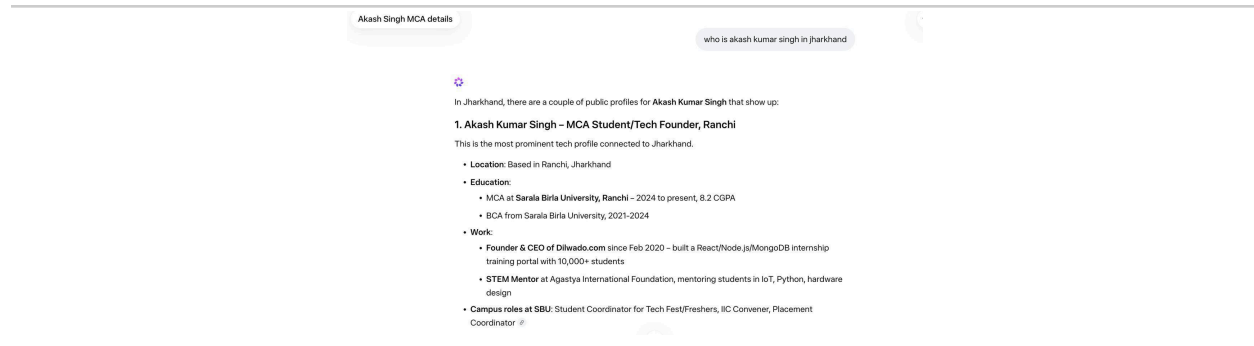


Figure 13. Perplexity AI response for 'who is akash kumar singh in ranchi' (April 2026). Source chip indicates four distinct domains cited. The subject is identified as the primary 'Akash Kumar Singh' result for the Ranchi area, suggesting entity disambiguation extends to location-qualified queries within the tested sample. Screenshot retained by author.

For the query 'who is akash kumar singh tech', Perplexity displayed a photo carousel sourced from the Netflix domain the only photo carousel in any AI response in the dataset. For the geographic query variant 'in ranchi', the source chip showed four distinct domains cited, and the subject was identified as the primary 'Akash Kumar Singh' result for the Ranchi area.

4.3.6 Meta AI 'who is akash kumar singh in jharkhand' (Figure 14)

The Meta AI response for the Jharkhand query frames the subject as the prominent tech profile connected to Jharkhand (population approximately 38 million). Every data point, including CGPA 8.2, founding date, tech stack, STEM Mentor role, and all three campus positions, originates from the GEO-optimised structured profiles. [Screenshot retained by author; Figure 14 available upon request.]

The pattern across all four platforms is consistent with the theoretical distinction between RAG and parametric architectures identified by Sharma (2026). This three-way pattern RAG-granular, hybrid-comprehensive, parametric-summary has not, to the authors' knowledge, been previously documented using a single controlled entity across all four platforms simultaneously.

4.4 Quantitative Summary Tables and Platform Comparison

Tables 4 and 5 present a before-versus-after comparison and a query-by-platform response granularity scoring. The granularity bar chart (Figure 15) visualises the mean scores.

Table 4

Before-vs.-After Comparison: Entity-Aware Result Counts Across Observation Surfaces

Observation Surface	Before Interventions (Informal; n = 0 logged)	After Interventions April 2026 (n = 14 screenshots)
Google Search	0 entity-specific results for subject; multiple unrelated individuals returned	4 consecutive entity-specific results (positions 2–5): LinkedIn,

Observation Surface	Before Interventions (Informal; n = 0 logged)	After Interventions April 2026 (n = 14 screenshots)
		Dilwado.com, 2 × SBU Facebook. Documented in Figure 1.
Google Images	0 entity-specific filter chips; no stable photograph attributed to subject	1 identity filter chip ('Tech entrepreneur'); 3 independent entity nodes; cross-platform photo associations across ≥6 sources. Documented in Figures 2–7.
Perplexity AI (3 query variants)	0 entity-aware responses retrieved	3/3 queries returned structured entity profiles; CGPA, tech stack, and campus roles accurately retrieved; photo carousel in 1/3 queries. Figures 8, 12, 13.
Google Gemini (2 query variants)	0 entity-aware responses observed	2/2 queries returned rich entity profiles including campus leadership, institutional events, and YouTube metrics. Figures 9–10.
Meta AI (2 query variants)	0 relevant responses observed	2/2 queries returned granular entity profiles; tech stack, competition details, Agile training data accurately retrieved; Netflix domain cited. Figures 11, 14.
ChatGPT GPT-4o (1 query variant)	0 entity-aware responses observed	1/1 returned accurate summary-level response: SBU, Dilwado.com, Google Student Ambassador role. Less granular than RAG-based platforms.

Note. Pre-intervention observations are informal and not formally logged; they should be interpreted as indicative, not definitive. Post-intervention observations are documented through fourteen original screenshots retained by the author. Results reflect the April 2026 observation window and are subject to change as AI systems are updated.

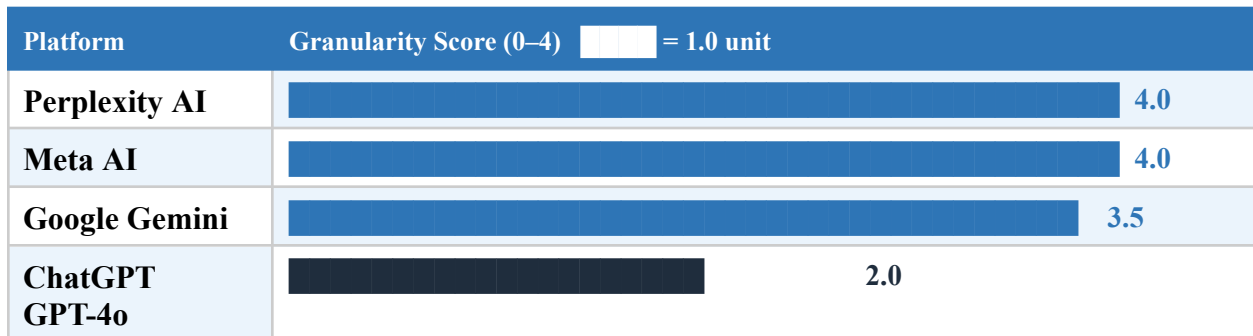
Table 5
Query-by-Platform Response Granularity Scoring (Ordinal Scale 0–4)

Query	Perplexity (RAG)	Meta AI (Hybrid)	Gemini (Hybrid)	ChatGPT (Parametric)
'mca student akash kumar singh'	4			2
'akash kumar singh tech'	4	4		

Query	Perplexity (RAG)	Meta AI (Hybrid)	Gemini (Hybrid)	ChatGPT (Parametric)
'akash kumar singh sbu'			4	
'akash kumar singh entrepreneur'			3	
'akash kumar singh in ranchi'	4			
'akash kumar singh in jharkhand'		4		
Mean Score (tested queries only)	4.0	4.0	3.5	2.0

Note. Scoring rubric: 0 = No entity-aware response; 1 = Name recognised, no specific facts; 2 = Correct name + institution + startup; 3 = Adds role details/achievements; 4 = Structured entity profile with specific verifiable facts (CGPA, founding date, tech stack, campus roles). Scores assigned by a single rater (the author); treat as indicative pending independent verification. '' = Query not tested on this platform.

Figure 15. Comparison of Mean Response Granularity Scores by AI Platform (Ordinal Scale 0–4). Perplexity AI (RAG) and Meta AI (hybrid) achieved the highest mean scores (4.0), followed by Google Gemini (3.5, hybrid) and ChatGPT (2.0, primarily parametric). Scores are based on author assessment of screenshot content; see Table 5 for rubric. $n = 1$ per platform; error bars not applicable.



Note. Each  block represents 0.1 granularity units on the 0–4 ordinal scale. Scores are author-assigned; see rubric in Table 5.

5. A Three-Stage SEO-GEO Integration Model

The intervention sequence and observed outcomes are consistent with a coherent three-stage process for building AI search visibility under the resource constraints typical of Indian student startup founders. The model is proposed as a descriptive and generative framework, grounded in both the observed case and the existing GEO literature (see Figure 16). It is not claimed as a validated causal theory; its value lies in providing a structured starting point for future research and for practitioners seeking a reproducible intervention sequence.

Figure 16. *Three-Stage SEO-GEO Integration Model. Stage 1 (Foundational SEO Presence) establishes a coherent cluster of indexable, fact-dense properties. Stage 2 (Entity Consolidation) aligns signals across independent domains to enable Knowledge Graph entity merging. Stage 3 (Earned Authority Signals) introduces third-party institutional corroboration to strengthen AI retrieval credibility. Stages are sequential and cumulative.*

Stage 1	Stage 2	Stage 3
Foundational SEO Presence • LinkedIn optimisation • Startup domain entity page • Static portfolio (identity-phrase slug) • Fact-dense content on each property	Entity Consolidation • Uniform role nomenclature • Consistent professional photograph • Domain-slug strategy • Cross-platform attribute alignment	Earned Authority Signals • Official university announcements • Independently indexed content • Verified, real-world achievements • Third-party institutional corroboration
<i>Prerequisite: none</i>	<i>Prerequisite: Stage 1 complete</i>	<i>Prerequisite: Stages 1 & 2 complete</i>

Note. Stages are sequential; Stage 1 must be completed before Stages 2 and 3 can take effect. Stage 3 is not under the direct control of the practitioner it depends on institutional recognition of real, verifiable achievements.

Stage 1: Foundational SEO Presence

The first stage involves establishing a small set of clearly indexable, keyword-aligned digital properties an optimised LinkedIn profile, a founder entity page on the startup domain, and a static portfolio site with an identity-phrase domain slug. The unifying design principle is coherence: every property presents the same core identity information (name, institution, startup role, key achievements) in language that maps directly onto target queries. This stage aligns with Sharma's (2026) finding that foundational SEO signals remain associated with Perplexity visibility. It is necessary but not sufficient for AI search visibility; its completion is a prerequisite for Stages 2 and 3.

Stage 2: Cross-Platform Entity Consolidation

The second stage involves deliberately aligning signals across independently hosted properties so that search and AI systems can resolve multiple references to a single coherent entity record. This requires uniform role nomenclature, consistent institutional naming, and a single professional photograph used consistently across all properties. The domain-slug strategy extends this alignment to the URL level. Evidence consistent with this stage is visible in Figures 2 through 7: a 'Tech entrepreneur' filter chip was generated using the exact role language deployed consistently across three domains; three independent entity nodes were established; and cross-platform photo associations extended beyond the three optimised properties. These observations are consistent with Volpini et al.'s (2026) entity memory layer framework and Noy et al.'s (2019) characterisation of consistent attribute naming as a material input to Knowledge Graph construction.

Stage 3: Earned Authority Signals

The third stage involves securing earned media from independent, authoritative sources. In this case, two official Sarala Birla University Facebook posts each arising from real, independently verifiable events, published by the university without request or payment appeared in Google Search results and contributed specific facts to AI assistant responses across all four platforms. Chen et al. (2025) proposed that AI search systems weight earned, authoritative third-party content more heavily than brand-owned material a pattern broadly consistent with the observations here.

The potentially novel observation of this study is the identification of official university communication channels as an accessible earned media pathway for student founders. Aggarwal et al. (2024) and Chen et al. (2025) discuss earned media primarily in terms of press coverage and news citations, which are structurally inaccessible to most student entrepreneurs. This study suggests that institutional recognition events ambassador selections, hackathon results, conference speaking invitations when officially announced on indexed institutional channels, may function as GEO authority signals with measurable outcomes within the tested sample. This remains a hypothesis requiring multi-case confirmation.

6. Discussion

6.1 Theoretical Implications

The findings of this study are broadly consistent with the existing GEO literature while suggesting extensions in two specific directions. First, they are consistent with the three core GEO principles established by Aggarwal et al. (2024) structured content, fact density, and earned media authority applying recognisably in a non-Western, zero-budget, common-name context. Second, they provide a direct platform-to-platform comparison of RAG versus parametric response granularity using a single controlled entity, illustrating empirically the architectural distinction that Sharma (2026) identifies in aggregate data.

The observation that Gemini retrieves content from beyond the directly optimised properties including institutional event pages and YouTube channel data carries a practically important implication: AI entity representation appears not to be bounded by what a founder deliberately optimises. Any publicly documented institutional activity involving the founder may contribute to their AI search profile. This suggests that monitoring tools designed to track AI response content should scan across a wider range of indexed content than just the properties under direct control.

The finding that RAG-based platforms reproduce specific, verifiable facts with notable fidelity while parametric platforms produce accurate but less granular summaries (mean scores of 4.0 vs. 2.0 on the ordinal scale; Table 5) has direct implications for GEO content strategy. Founders seeking to appear in RAG-based systems may benefit from prioritising structured, fact-dense entity pages over narrative content. Founders seeking to appear in parametric systems may benefit more from widespread indexed presence across multiple independent platforms.

6.2 Practical Implications for Indian Student Founders

Three provisional observations emerge from this study, framed as tentative guidance pending multi-case confirmation:

- **Concentrate effort on a few well-maintained, fact-dense properties.** Three properties appear sufficient to establish Stage 1 foundational presence within the tested sample, provided they consistently carry the same structured identity information.
- **Treat role nomenclature and institutional naming as GEO signals, not cosmetic choices.** The exact phrase used in a LinkedIn headline may appear in AI-generated answers. Inconsistency across platforms may undermine entity consolidation.
- **Treat institutional recognition events as potential GEO assets.** Ambassador selections, hackathon results, government scheme approvals, and conference speaking invitations when announced on publicly indexed institutional channels may function as earned media signals contributing to AI assistant response accuracy.

6.3 Scope and Generalisability

What this study establishes, with documented evidence, is that a specific sequence of zero-budget interventions was associated with a specific pattern of AI search visibility outcomes for one subject, in one context, over one observation window. It does not establish whether this pattern holds for other subjects, other names, other institutions, other countries, or other observation windows. It does not establish causation. It cannot control for concurrent changes in AI systems during the intervention period. These are real limitations that define the scope of the contribution not arguments against its value at the appropriate level of exploratory, hypothesis-generating inquiry.

7. Limitations

The following limitations are inherent to the design and should be considered when interpreting the findings.

- **Single-case design.** The study examines one individual, one startup, and a bounded set of queries. No claim about the frequency or generalisability of these outcomes is supported by the evidence.
- **Researcher-as-subject.** The individual who designed, implemented, and observed the interventions is also the subject of study. No human participants other than the author were involved. This role overlap introduces potential for confirmation bias. Multi-device, multi-account observation and retained screenshots partially mitigate this risk but cannot eliminate it.
- **Absence of formal baseline logging.** Pre-intervention states were observed informally and not logged. This is the most consequential methodological gap: a formally logged baseline would substantially strengthen any causal inference drawn from the results.
- **Absence of a control case.** The study does not include a parallel individual sharing the subject's name and context who did not implement interventions. Without a control, it is not possible to determine how much of the observed change reflects the interventions rather than concurrent changes in AI systems.
- **Personalisation effects.** Search personalisation driven by account history, cookies, and device type was not fully controlled. Some results may reflect prior search behaviour rather than a pattern generalised to other users.
- **Temporal specificity.** AI model parameters, retrieval indices, and ranking algorithms are updated continuously. Results documented here reflect one observation window in April 2026 and cannot be assumed to persist.
- **Language scope.** All queries were issued in English. GEO dynamics for Hindi-language or regional-language queries are not addressed.
- **Hallucination rate not formally measured.** Factual accuracy was assessed by visual inspection and cross-referencing with publicly verifiable information. No systematic hallucination-rate measurement was conducted across repeated queries.
- **Unverified metrics.** Statistics such as 10,000+ registered students, 400,000 YouTube subscribers, and 13.7 million total views are drawn from the author's public profiles and have not been independently verified. They are reported as author-reported or platform-reported estimates.
- **Single-rater granularity scoring.** The ordinal granularity scores in Table 5 were assigned by the author without independent inter-rater verification. They should be treated as indicative rather than definitive quantitative measures pending replication with independent raters.

8. Conclusion

This paper has presented what is, to the authors' knowledge, among the first screenshot-evidenced case studies of GEO implementation by a student entrepreneur at a regional Indian university. A structured sequence of zero-budget interventions foundational SEO presence, cross-platform entity consolidation, and institutionally earned media was associated with documentable improvements in AI search visibility across Google Search, Google Images, and four major AI assistant platforms within the tested sample. The fourteen evidence screenshots, together with the quantitative granularity scoring (Table 5) and before-versus-after comparison (Table 4), constitute a substantive evidentiary record for a single-case qualitative study in this nascent field.

The study makes two contributions suggesting extensions to the existing GEO literature. First, the observations are consistent with GEO principles established by Aggarwal et al. (2024) and Chen et al. (2025) applying recognisably in a zero-budget, common-name, regional university context. Second, official university communication channels appear to represent an accessible and potentially effective GEO earned media pathway, providing a practical alternative to press coverage for student founders without media access.

The three-stage SEO-GEO Integration Model Foundational Presence, Entity Consolidation, Earned Authority Signals is offered as a generative framework for future research and practitioner guidance, not as a confirmed causal theory. Multi-case confirmation across diverse naming contexts, institution types, and geographic regions, with formal control entities and systematic baseline logging, is the most important direction for future work. A longitudinal component tracking AI response stability over six to twelve months would address the temporal specificity limitation. Development of a lightweight entity monitoring tool regularly querying AI platforms for a startup's key identity phrases and recording changes over time would provide founders with practical feedback infrastructure for managing GEO as an ongoing activity.

Declarations

Funding

This research received no external funding. All interventions were implemented at zero monetary cost by the author.

Conflicts of Interest

The author declares no conflicts of interest beyond the researcher-as-subject role disclosed in the Author Note and Limitations section. No financial relationships exist with any organisations mentioned in this paper.

Ethical Statement

This study analysed publicly available digital content generated by the author across platforms to which the author has provided consent. No personal data of third parties were collected, processed, or reported. No human participants other than the author were involved in any aspect of data generation or collection. No vulnerable populations were involved. No institutional ethics approval was required under applicable guidelines. Findings are presented as exploratory and hypothesis-generating in nature.

Data Availability

The fourteen screenshots constituting the primary evidence are retained by the author and are available to reviewers and editors upon request. No other data were generated for this study. See Appendix A for a full data source map.

Author Contributions

Akash Kumar Singh: Conceptualisation, Methodology, Investigation, Data Curation, Formal Analysis, Writing – Original Draft, Writing – Review & Editing.

Acknowledgements

The author thanks the faculty and administration of Sarala Birla University, Ranchi, for their institutional support and for the official recognition events whose documented announcements form part of the empirical evidence in this study.

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Note. References marked with † are arXiv preprints and have not undergone formal peer review.

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Appendix A: Data Source Map and Claim Verification

Table A1 maps each factual claim in the paper to its source, verification status, and figure reference. This appendix is provided in the interest of full transparency regarding the evidentiary basis of all assertions.

Table A1
Data Source Map: Claim-to-Evidence Mapping

Claim	Source / Evidence	Verification Status	Figure / Table Ref.
MCA student at SBU, 2024–present; CGPA 8.2	Author's LinkedIn profile; Perplexity AI response screenshot	Author-reported; visible in indexed public profile	Figs 8, 11; Table 2
BCA from SBU, 2021–2024	Author's LinkedIn profile; AI response screenshots	Author-reported; consistent across multiple AI responses	Figs 8, 11
Dilwado.com founded Feb 2020; AICTE-approved; 10,000+ students	Dilwado.com entity page; Perplexity and Meta AI screenshots	Founding date & AICTE status: author-reported and publicly listed. Student count: author-reported estimate, not independently audited.	Figs 8, 11; Table 2
YouTube: 400,000+ subscribers; 13.7M views	Public YouTube channel (publicly viewable at time of observation); Google Gemini screenshots	Platform-reported figures visible on public channel page; not independently audited by this study.	Fig 10; Table 3
Google Student Ambassador selection	Official SBU Facebook post (40+ reactions, indexed ~7 months prior); Google Search result snippet (Fig 1)	Independently posted by SBU; publicly indexed; verifiable via Facebook URL.	Fig 1; Table 2
Team AIRAVATA hackathon leadership	Official SBU Facebook post (50+ reactions, indexed ~7 months prior); Google Search result snippet (Fig 1)	Independently posted by SBU; publicly indexed; verifiable via Facebook URL.	Fig 1; Table 2

Claim	Source / Evidence	Verification Status	Figure / Table Ref.
STEM Mentor at Agastya International Foundation (since Dec 2025)	Author's LinkedIn profile; Perplexity AI response (Fig 8)	Author-reported; reproduced in AI response screenshot.	Fig 8; Table 2
300+ students in Agile training programmes	Author's Netlify portfolio (cited by Meta AI, Fig 11)	Author-reported estimate on GEO-optimised profile; not independently audited.	Fig 11; Table 3
Tech Pragati 2026 organiser; ICACA-2026 speaker	Google Gemini responses (Figs 9–10); institutional event pages	Attributed by Gemini, suggesting retrieval from non-optimised institutional sources. Verifiable via SBU event records.	Figs 9, 10
Organic CTR fell ~65% on AI Overview queries	Seer Interactive (2025), reported in Search Engine Land	Industry report; methodology not peer-reviewed. Cited as indicative.	Section 1
GEO increases citation rates up to 40%	Aggarwal et al. (2024) peer-reviewed, ACM SIGKDD	Peer-reviewed; controlled experimental finding.	Sections 1, 2.1, 5
Referring domains: $r = 0.319$, $p < 0.001$ with Perplexity visibility	Sharma (2026) arXiv preprint	arXiv preprint; not yet peer-reviewed. Cited with preprint flag.	Sections 1, 2.1, 4.3

Note. All author-reported metrics are drawn from the subject's publicly indexed profiles and represent self-reported or platform-reported figures. Independent audit was not conducted as part of this study. Reviewers wishing to verify any claim are invited to contact the author for screenshot access.

Appendix B: Screenshot Evidence Index

Table B1 provides a complete index of all fourteen screenshots constituting the primary empirical evidence of this study. Screenshots are retained by the author in their original form and are available to reviewers upon request.

Table B1
Complete Screenshot Evidence Index (n = 14)

Fig.	Platform / Surface	Query	Key Evidence	File Reference
1	Google Search	<i>mca student akash kumar singh</i>	4-result entity cluster (positions 2–5)	wa_1.jpeg
2	Google Images	<i>akash kumar singh</i>	'Tech entrepreneur' filter chip appears first	evidence_2.png
3	Google Images	<i>akash kumar singh tech</i>	3-domain cluster in top positions	evidence_3.png
4	Google Images	<i>akash kumar singh tech</i>	LinkedIn entity node with cross-platform associations	evidence_4.png
5	Google Images	<i>akash kumar singh tech</i>	Netlify portfolio named entity node	evidence_5.png
6	Google Images	<i>akash kumar singh tech</i>	Dilwado.com entity node	evidence_6.png
7	Google Images	<i>akash kumar singh tech</i>	Cross-platform photo associations (4 platform sources)	evidence_7.png
8	Perplexity AI	<i>who is akash kumar singh mca</i>	Structured entity profile; CGPA 8.2; all roles retrieved	wa_2.jpeg
9	Google Gemini	<i>who is akash kumar singh sbu</i>	Tech Pragati 2026 & ICACA-2026 retrieved from non-optimised sources	wa_3.jpeg
10	Google Gemini	<i>who is akash kumar singh entrepreneur</i>	YouTube 400K+ subscribers	wa_4.jpeg

Fig.	Platform / Surface	Query	Key Evidence	File Reference
			retrieved from non-optimised source	
11	Meta AI	<i>who is akash kumar singh tech</i>	Anveshna detail & Agile 300+ students; Netlify cited	wa_5.jpeg
12	Perplexity AI	<i>who is akash kumar singh tech</i>	Photo carousel from Netlify domain (unique in dataset)	wa_6.jpeg
13	Perplexity AI	<i>who is akash kumar singh in ranchi</i>	4 domains cited; subject identified as primary Ranchi result	wa_7.jpeg
14	Meta AI	<i>who is akash kumar singh in jharkhand</i>	All facts from GEO-optimised profiles; state-level prominence	[retained by author]

Appendix C: GEO Practitioner Checklist for Indian Student Founders

The following checklist is derived from the three-stage model proposed in Section 5 and from the intervention sequence documented in this study. It is offered as a tentative practical guide for student founders seeking to replicate the intervention approach under comparable resource constraints. The checklist is presented as a hypothesis-generating tool, not a validated prescriptive framework. All items should be verified against current platform policies and AI system behaviours, which are subject to continuous change.

Stage 1: Foundational SEO Presence

LinkedIn Profile

- ☐ **URL slug:** Set your LinkedIn URL to include your full name and a key role phrase (e.g., linkedin.com/in/yourname-tech-entrepreneur)
- ☐ **Headline:** Include: [Degree] Student | [Role] | [Company] ([Accreditation if applicable])
- ☐ **About section:** Open with a structured identity statement: full name, university name and location, degree and year, company name and founding date, core metric (students served, etc.)
- ☐ **Experience:** Each role includes: exact title, company name as it appears on the company domain, start date, and one verifiable metric
- ☐ **Professional photograph:** Upload a clear, consistent photograph that will be used identically across all three properties

Startup Domain Entity Page

- ☐ **URL path:** Create a long-form page at yourdomain.com/about/ or /founder/ with your name and role phrase in the URL
- ☐ **H1 heading:** Title the page with your identity phrase (e.g., 'Who is [Your Name] Tech?')
- ☐ **Fact-dense body:** Include: university name + location, degree + year, CGPA (if relevant), company name + founding date + accreditation status + student/user count, tech stack, and all institutional recognitions with approximate dates
- ☐ **Same photograph:** Use the identical professional photograph as on LinkedIn

Static Portfolio Site

- ☐ **Domain slug:** Create a site at [yourname]-[role]-tech.[host].app (e.g., yourname-tech-entrepreneur.netlify.app)
- ☐ **Content mirror:** Mirror the core identity facts from the entity page, using the same language and role phrases
- ☐ **Same photograph:** Use the identical professional photograph

- ☐ **Indexing:** Verify the site is indexed by submitting to Google Search Console; allow 2–4 weeks for full indexing

Stage 2: Entity Consolidation

- ☐ **Uniform role nomenclature:** Choose one canonical version of each role title and use it identically across all three properties. Do not vary capitalisation, punctuation, or wording.
- ☐ **Uniform university naming:** Use the official full name of your university consistently (e.g., 'Sarala Birla University' not 'SBU')
- ☐ **Uniform company naming:** Use the name exactly as registered and as it appears on the company's primary domain
- ☐ **Cross-check:** Audit all three properties side by side; identify and eliminate any discrepancies in name, title, institution, or date formats
- ☐ **Allow indexing time:** Allow at least 4–8 weeks for all three properties to be fully indexed before expecting AI retrieval effects

Stage 3: Earned Authority Signals

- ☐ **Real achievements:** Pursue verifiable, officially recognised achievements: ambassador programme selection, hackathon prizes, government scheme approvals, conference speaking invitations
- ☐ **Request official announcements:** After each achievement, request the relevant institutional contact (university communications office, programme coordinator) to publish an official announcement on an indexed institutional channel (university website, verified social media account)
- ☐ **Verify indexing:** Search for the announcement by name to verify it has been indexed by Google. If not indexed within 4 weeks, request submission via Google Search Console.
- ☐ **Name consistency in announcements:** Verify that the announcement uses your name exactly as it appears on your three optimised properties inconsistency may reduce entity merging probability

Stage 4 (Ongoing): Monitor and Maintain

Note. Stage 4 is not part of the three-stage model but is recommended as ongoing practice following initial implementation.

- ☐ **Monthly AI query check:** Query all four platforms (Perplexity, Gemini, Meta AI, ChatGPT) with your role-qualified name and document the response for comparison over time
- ☐ **Fact accuracy verification:** Check that AI-generated responses cite accurate facts. If inaccurate information appears, update the relevant entity page with corrected, more prominent facts.

- ☐ **Platform nomination check:** Verify that AI responses still cite your preferred domain as a primary source. If Netlify or LinkedIn drops out, check that the relevant property remains indexed and fact-dense.
- ☐ **New achievements:** Update all three properties simultaneously when new verifiable achievements occur. Pursue official announcements for each significant event.

Important Caveats

This checklist is derived from a single case study and has not been validated across multiple subjects, institutions, or geographic regions. AI system behaviours are subject to continuous change; what produces AI retrieval effects in April 2026 may not produce the same effects at a later date. All items should be treated as tentative guidance rather than proven prescriptions. Practitioners are advised to document their own interventions and outcomes systematically, contributing to the multi-case evidence base that this field requires.